Poisson方程的$C^{2,\alpha}$估计*

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♣ **写在前面.**

对于基本解我们有如下估计：

$$|D\Gamma(x-y)| \leq \frac{C}{|x-y|^{n-1}}, \quad |D^2\Gamma(x-y)| \leq \frac{C}{|x-y|^n}.$$

从而我们可写为

$$|D\Gamma| = O(|x-y|^{1-n}), \quad |D^2\Gamma| = O(|x-y|^{-n}).$$

下面给出引理（参见[2]17.5命题4）

引理 1 考虑含参变量$x \in \Omega$的积分

$$F(x) = \int_{\Omega'} f(x,y)dy,$$

若f满足估计式

$$|f(x,y)| \leq \frac{M}{|x-y|^\alpha},$$

其中$M \in \mathbb{R}$，$\alpha < n$，则积分在Ω上一致收敛.

因此对于积分

$$F(x) = \int_\Omega |D^2\Gamma(x-y)|dy$$

一般是不可积的.这也就解释了我们即将介绍的一部分内容不能平凡的推广.

考虑Poisson方程解的正则性.

♣ **讨论Poisson方程的一个特殊解的正则性.**

定义 1 (Newton位势) 设Ω为一个有界区域，f为一个有界函数，定义

$$w(x) = \int_\Omega \Gamma(x-y)f(y)dy,$$

其中 $\Gamma(x-y) = \begin{cases} \dfrac{1}{2\pi}\log|x-y| & n=2 \\ \dfrac{1}{(2-n)\omega_n}|x-y|^{2-n} & n \geq 3 \end{cases}$.则称$w$为$f$在$\Omega$上的*Newton*位势.

注记 1 需要注意的是虽然$\Gamma(x-y)$在$y=x$处存在奇点，但$w(x)$仍是*well-defined*.事实上，根据**引理*1***得知.

*椭圆方程讨论班

注记 2 后面即将说明*Newton*位势为*Poisson*方程$\Delta u = f$的解.

接下来讨论Newton位势的正则性.

• C^1**正则性.**

命题 1 设$\Omega = B_1$为$\mathbb{R}^n$上的单位球，f在B_1上有界，w为f在B_1上的*Newton*位势，则$w \in C^1(\mathbb{R}^n)$.对任意的$x \in \mathbb{R}^n$，$i = 1, 2, \cdots, n$，有

$$D_i w = \int_{B_1} D_{x_i}\Gamma(x-y)f(y)dy.$$

进一步，有估计：

$$|w|_{C^1(\overline{B}_1)} \le C|f|_{L^\infty(B_1)} \quad (C \sim n).$$

证明 令

$$v_i = \int_{B_1} D_i\Gamma(x-y)f(y)dy.$$

由于$|D\Gamma| = \begin{cases} O(|x-y|^{-1}) & n=2 \\ O(|x-y|^{1-n}) & n \ge 3 \end{cases}$，又$f(y)$为有界函数，则$v_i(x)$在$\mathbb{R}^n$上well-defined，且连续.

选取$\eta \in C^1(\mathbb{R}$，$0 \le \eta \le 1$，$0 \le \eta' \le 2$，$\eta = \begin{cases} 0 & t \le 1 \\ 1 & t \ge 2 \end{cases}$.考虑（$\varepsilon > 0$）

$$w_\varepsilon = \int_{B_1} \Gamma\eta_\varepsilon f(y)dy,$$

其中$\eta_\varepsilon = \eta\left(\dfrac{|x-y|}{\varepsilon}\right)$. 则$\Gamma\eta_\varepsilon \in C^1(\mathbb{R}^n)$，从而$w_\varepsilon(x) \in C^1(\mathbb{R}^n)$.
♠ 计算.

$$\begin{aligned}
|w(x) - w_\varepsilon(x)| &= \left|\int_{B_1} \Gamma(x-y)f(y)dy - \int_{B_1} \Gamma(x-y)\eta_\varepsilon f(y)dy\right| \\
&\le \int_{B_1} |\Gamma(x-y)(1-\eta_\varepsilon)f(y)|\,dy \\
&\le \sup|f| \cdot \int_{B_{2\varepsilon}(x)} |\Gamma(x-y)|dy \\
&= \sup|f| \cdot \begin{cases} \int_{B_{2\varepsilon}(x)} \left|\dfrac{1}{2\pi}\log|x-y|\right| dy & n=2 \\ \int_{B_{2\varepsilon}(x)} \left|\dfrac{1}{(2-n)\omega_n}|x-y|^{2-n}\right| dy & n \ge 3 \end{cases}
\end{aligned}$$

其中第三个不等号是由于，1.在$|x-y| \ge 2\varepsilon$处，$\eta_\varepsilon = \eta(t) = 0$（$t \ge 2$）;2.$0 \le \eta \le 1$.
分别计算.

$$\begin{aligned}
\int_{B_{2\varepsilon}(x)} \left|\frac{1}{2\pi}\log|x-y|\right| dy &= \int_{B_{2\varepsilon}(x)} -\frac{1}{2\pi}\log|x-y|dy \\
&= -\int_0^{2\varepsilon} \frac{1}{2\pi}\log r \cdot 2\pi \cdot r dr \\
&= -\int_0^{2\varepsilon} \log r \cdot r dr \\
&\le -\log(2\varepsilon)(2\varepsilon)^2.
\end{aligned}$$

其中最后一个不等号是由$-\log r\cdot r$在$(0,2\varepsilon)$上单调递增.

$$\begin{aligned}\int_{B_{2\varepsilon}(x)}\left|\frac{1}{(2-n)\omega_n}|x-y|^{2-n}\right|dy=&-\int_{B_{2\varepsilon}(x)}\frac{1}{(2-n)\omega_n}|x-y|^{2-n}dy\\=&-\int_0^{2\varepsilon}\frac{1}{(2-n)\omega_n}\cdot r^{2-n}\cdot\omega_n r^{n-1}dr\\=&-\int_0^{2\varepsilon}\frac{1}{2-n}rdr\\\leq&(2\varepsilon)^2.\end{aligned}$$

注意上面的计算中我们应用了如下事实：

$$\begin{aligned}&\int_0^r\left(\int_{\partial B_s(x)}u(y)dS_y\right)ds=\int_{B_r(x)}u(y)dy\\\Longrightarrow&\int_0^{2\varepsilon}\left(\int_{\partial B_r(x)}u(y)dS_y\right)dr=\int_{B_{2\varepsilon}(x)}u(y)dy\\\Longrightarrow&\int_0^{2\varepsilon}\omega_n\cdot r^{n-1}dr=\int_{B_{2\varepsilon}(x)}u(y)dy.\end{aligned}$$

因此

$$|w(x)-w_\varepsilon(x)|\leq\begin{cases}-\log(2\varepsilon)(2\varepsilon)^2\sup|f| & n=2\\(2\varepsilon)^2\sup|f| & n\geq 3\end{cases}.$$

从而

$$w_\varepsilon(x)\longrightarrow w(x)\quad\varepsilon\longrightarrow 0,$$

即$w_\varepsilon(x)$在$\mathbb{R}^n$上一致收敛到$w(x)$. 由于$w_\varepsilon(x)\in C^1(\mathbb{R}^n)$，则$w(x)\in C^1(\mathbb{R}^n)$.

♠ 计算.

$$\begin{aligned}|v_i(x)-D_iw_\varepsilon(x)|=&\left|\int_{B_1}D_i\Gamma(x-y)f(y)dy-D_i(\Gamma(x-y)\eta_\varepsilon)f(y)dy\right|\\=&\left|\int_{B_1}D_i((1-\eta_\varepsilon)\Gamma(x-y))f(y)dy\right|\\\leq&\sup|f|\int_{B_{2\varepsilon}(x)}|D_i((1-\eta_\varepsilon)\Gamma(x-y))|\,dy\\\leq&\sup|f|\int_{B_{2\varepsilon}(x)}|-D_i\eta_\varepsilon\cdot\Gamma(x-y)|+|(1-\eta_\varepsilon)D_i\Gamma(x-y)|dy\\\leq&\sup|f|\int_{B_{2\varepsilon}(x)}\left|-\frac{2}{\varepsilon}\Gamma(x-y)\right|+|D_i\Gamma(x-y)|dy\\=&\sup|f|\int_{B_{2\varepsilon}(x)}\left(\frac{2}{\varepsilon}|\Gamma|+|D_i\Gamma|\right)dy\\=&\sup|f|\cdot\begin{cases}\int_{B_{2\varepsilon}(x)}\left(\frac{2}{\varepsilon}\cdot\left|\frac{1}{2\pi}\log|x-y|\right|+\left|D_i\left(\frac{1}{2\pi}\log|x-y|\right)\right|\right)dy & n=2\\\int_{B_{2\varepsilon}(x)}\left(\frac{2}{\varepsilon}\cdot\left|\frac{1}{(2-n)\omega_n}|x-y|^{2-n}\right|+\left|D_i\left(\frac{1}{(2-n)\omega_n}|x-y|^{2-n}\right)\right|\right)dy & n\geq 3\end{cases}\end{aligned}$$

分别计算.

$$
\begin{aligned}
&\int_{B_{2\varepsilon}(x)} \left(\frac{2}{\varepsilon} \left| \frac{1}{2\pi} \log|x-y| \right| + \left| D_i \left(\frac{1}{2\pi} \log|x-y| \right) \right| \right) dy \\
=&\int_0^{2\varepsilon} \left(\frac{2}{\varepsilon} \left| \frac{1}{2\pi} \log r \right| + \left| D_i \left(\frac{1}{2\pi} \log r \right) \right| \right) \cdot 2\pi \cdot r dr \\
=&\int_0^{2\varepsilon} \left(-\frac{2}{\varepsilon} \log r + |D_i(\log r)| \right) \cdot r dr \\
\leq&\int_0^{2\varepsilon} \left(-\frac{2}{\varepsilon} \log r + \frac{1}{r} \right) \cdot r dr \\
\leq& -8\log(2\varepsilon) \cdot \varepsilon + 2\varepsilon.
\end{aligned}
$$

$$
\begin{aligned}
&\int_{B_{2\varepsilon}(x)} \left(\frac{2}{\varepsilon} \cdot \left| \frac{1}{(2-n)\omega_n} |x-y|^{2-n} \right| + \left| D_i \left(\frac{1}{(2-n)\omega_n} |x-y|^{2-n} \right) \right| \right) dy \\
=&\int_0^{2\varepsilon} \left(\frac{2}{\varepsilon} \left| \frac{1}{(2-n)\omega_n} r^{2-n} \right| + \left| D_i \left(\frac{1}{(2-n)\omega_n} r^{2-n} \right) \right| \right) \cdot \omega_n \cdot r^{n-1} dr \\
=&\int_0^{2\varepsilon} \left(-\frac{2}{\varepsilon} \cdot \frac{1}{2-n} \cdot r + \left| D_i \left(\frac{1}{2-n} r^{2-n} \right) \right| \cdot r^{n-1} \right) dr \\
\leq&\int_0^{2\varepsilon} -\frac{2}{\varepsilon} \cdot \frac{1}{2-n} \cdot r dr + \int_0^{2\varepsilon} r^{1-n} \cdot r^{n-1} dr \\
=& -\frac{2}{\varepsilon} \cdot \frac{1}{2-n} \cdot \left(\frac{1}{2} r^2 \right) \Big|_0^{2\varepsilon} + 2\varepsilon \\
=&\frac{1}{n-2} \cdot 4\varepsilon + 2\varepsilon \\
\leq&6\varepsilon.
\end{aligned}
$$

因此

$$
|v_i(x) - D_i w_\varepsilon(x)| \leq \begin{cases} (-8\varepsilon\log(2\varepsilon) + 2\varepsilon) \sup|f| & n = 2 \\ 6\varepsilon \cdot \sup|f| & n \geq 3. \end{cases}
$$

从而

$$
D_i w_\varepsilon(x) \longrightarrow v_i(x) \quad \varepsilon \longrightarrow 0.
$$

即有：

$$
\begin{cases} w_\varepsilon(x) \longrightarrow w(x) \\ D_i w_\varepsilon(x) \longrightarrow v_i(x). \end{cases}
$$

故

$$
D_i w(x) = v_i(x).
$$

进一步给出估计：

$$
\begin{aligned}
|w|_{C^1(\overline{B}_1)} &= \sup_{\overline{B}_1} |w| + \sup_{\overline{B}_1} |Dw| \\
&= \sup_{\overline{B}_1} \left| \int_{B_1} \Gamma(x-y) f(y) dy \right| + \sup_{\overline{B}_1} \left| \int_{B_1} D\Gamma(x-y) f(y) dy \right| \\
&\leq C \cdot |f|_{L^\infty(B_1)}.
\end{aligned}
$$

□

• C^2**正则性.**

注记 3 我们需要注意的是，我们不能直接对D_iw求偏导，因为这时候$|D^2\Gamma(x-y)|$不一定可积.所以我们需要对f给一些条件.

命题 2 设$\Omega = B_1$为$\mathbb{R}^n$上的单位球，$f \in L^\infty(B_1) \cap C^\alpha(B_1)$（$\alpha \in (0,1)$），$w$为$f$在$B_1$上的*Newton*位势，则$w \in C^2(B_1)$.对任意的$x \in B_1$，$i,j = 1,2,\cdots,n$，有

$$D_{ij}w(x) = \int_{B_1} D_{x_ix_j}\Gamma(x-y)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_{x_j}\Gamma(x-y)\nu_i(y)dS_y.$$

证明 令

$$v_{ij} = \int_{B_1} D_{x_ix_j}\Gamma(x-y)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_{x_j}\Gamma(x-y)\nu_i(y)dS_y.$$

由于$|D\Gamma| = O(|x-y|^{1-n})$，$|D^2\Gamma| = O(|x-y|^{-n})$，$|f(y)-f(x)| = O(|x-y|^\alpha)$.则$v_{ij}(x)$为well-defined.
定义

$$v_{i,\varepsilon} = \int_{B_1} D_i\Gamma \cdot \eta_\varepsilon f(y)dy,$$

其中$\eta_\varepsilon = \eta\left(\dfrac{|x-y|}{\varepsilon}\right)$，$0 \le \eta \le 1$，$0 \le \eta' \le 2$，$\eta = \begin{cases} 0 & t \le 1 \\ 1 & t \ge 2. \end{cases}$ 由于$D_i\Gamma \cdot \eta_\varepsilon \in C^1(B)$，则$v_{i,\varepsilon} \in C^1(B_1)$.
♠ 计算.

$$\begin{aligned}
D_jv_{i,\varepsilon}(x) &= \int_{B_1} D_j(D_i\Gamma \cdot \eta_\varepsilon)f(y)dy \\
&= \int_{B_1} D_j(D_i\Gamma \cdot \eta_\varepsilon)(f(y)-f(x))dy + f(x)\int_{B_1} D_j(D_i\Gamma \cdot \eta_\varepsilon)dy \\
&= \int_{B_1} D_j(D_i\Gamma \cdot \eta_\varepsilon)(f(y)-f(x))dy - f(x)\int_{B_1} D_{y_j}(D_i\Gamma \cdot \eta_\varepsilon)dy \\
&= \int_{B_1} D_j(D_i\Gamma \cdot \eta_\varepsilon)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_i\Gamma \cdot \eta_\varepsilon \cdot \nu_j(y)dS_y \\
&= \int_{B_1} D_j(D_i\Gamma \cdot \eta_\varepsilon)(f(y)-f(x))dy - f(x)\int_{\partial B_1} D_i\Gamma \cdot \nu_j(y)dS_y
\end{aligned}$$

其中要最后一个等号成立，须保证ε充分小.
♠ 计算.

$$\begin{aligned}
|v_{ij}(x) - D_jv_{i,\varepsilon}(x)| &= \left|\int_{B_1} (D_{ij}\Gamma(x-y) - D_j(D_i\Gamma \cdot \eta_\varepsilon))(f(y)-f(x))dy\right| \\
&= \left|\int_{B_1} D_j((1-\eta_\varepsilon)D_i\Gamma)(f(y)-f(x))dy\right| \\
&\le [f]_\alpha \int_{B_{2\varepsilon}(x)} |D_j((1-\eta_\varepsilon)D_i\Gamma)| \cdot |y-x|^\alpha dy \\
&\le [f]_\alpha \int_{B_{2\varepsilon}(x)} \left(\frac{2}{\varepsilon}|D_i\Gamma| + |D_{ij}\Gamma|\right)|y-x|^\alpha dy \\
&\le [f]_\alpha \int_{B_{2\varepsilon}(x)} \left(\frac{2}{\varepsilon} \cdot |x-y|^{1-n} + |x-y|^{-n}\right)|y-x|^\alpha dy \\
&= [f]_\alpha \int_0^{2\varepsilon} \left(\frac{2}{\varepsilon} \cdot r^{1-n} + r^{-n}\right) r^\alpha \cdot \omega_n \cdot r^{n-1}dr
\end{aligned}$$

$$
\begin{aligned}
&=[f]_\alpha \cdot \omega_n \int_0^{2\varepsilon} \frac{2}{\varepsilon} \cdot r^\alpha + r^{\alpha-1} dr \\
&=[f]_\alpha \cdot \omega_n \cdot (2\varepsilon)^\alpha \left(\frac{4}{\alpha+1} + \frac{1}{\alpha} \right) \\
&\leq C \cdot (2\varepsilon)^\alpha.
\end{aligned}
$$

则

$$D_j v_{i,\varepsilon} \longrightarrow v_{ij}(x) \quad \varepsilon \longrightarrow 0,$$

即$D_j v_{i,\varepsilon}$在B_1的紧子集上一致收敛于v_{ij}.
由于

$$v_{i,\varepsilon}(x) \longrightarrow D_i w \quad \varepsilon \longrightarrow 0,$$

从而$w \in C^2(B_1)$，$D_{ij} w = v_{ij}(x)$. □

注记 4 这时我们能够说明*Newton*位势为*Poisson*方程的解.事实上，由于

$$D_{ij} w(x) = \int_{B_1} D_{x_i x_j} \Gamma(x-y)(f(y)-f(x))dy - f(x) \int_{\partial B_1} D_{x_j} \Gamma(x-y) \nu_i(y) dS_y$$

令$j=i$，再求和得

$$\Delta w(x) = \int_{B_1} \Delta_x \Gamma(x-y)(f(y)-f(x))dy + f(x) \int_{\partial B_1} \frac{\partial \Gamma}{\partial \nu_y}(x-y) dS_y$$

由于对$x \neq y$，有$\Delta_x \Gamma(x-y) = 0$，且$\int_{\partial B_1} \frac{\partial \Gamma}{\partial \nu_y}(x-y) dS_y = 1$，则

$$\Delta w(x) = f(x).$$

• $C^{2,\alpha}$**正则性.**

命题 3 设$\Omega = B_1$为$\mathbb{R}^n$上的单位球，$f \in L^\infty(B_1) \cap C^\alpha(\overline{B}_1)$（$\alpha \in (0,1)$），$w$为$f$在$B_1$上的*Newton*位势，则$w \in C^{2,\alpha}(B_1) \cap C(\overline{B}_1)$.进一步，有估计：

$$|w|_{C^{2,\alpha}(B_{1/2})} \leq C |f|_{C^\alpha(B_1)}.$$

证明 任取$x \in B_{1/2}$.
♠ 计算.

$$
\begin{aligned}
|D_{ij} w(x)| &= \left| \int_{B_1} D_{ij} \Gamma(x-y)(f(y)-f(x))dy - f(x) \int_{\partial B_1} D_i \Gamma(x-y) \cdot \nu_j(y) dS_y \right| \\
&\leq C \left(\int_{B_1} |x-y|^{-n} \cdot [f]_\alpha \cdot |x-y|^\alpha dy + |f| \int_{\partial B_1} |x-y|^{1-n} dS_y \right) \\
&= C \left([f]_\alpha \cdot \int_{B_1} |x-y|^{\alpha-n} dy + |f| \int_{\partial B_1} |x-y|^{1-n} dS_y \right) \\
&\leq C \left([f]_\alpha \int_0^1 r^{\alpha-n} \cdot \omega_n \cdot r^{1-n} dr + |f| \int_{\partial B_1} \left(\frac{1}{2} \right)^{n-1} dS_y \right) \\
&\leq C \left([f]_\alpha \omega_n \cdot \frac{1}{\alpha} + |f| \left(\frac{1}{2} \right)^{1-n} |\partial B_1| \right) \\
&\leq C([f]_\alpha + |f|).
\end{aligned}
$$

取$\bar{x} \in B_{1/2}$，令$r = |x - \bar{x}|$，我们需要证明

$$\frac{|D_{ij}w(x) - D_{ij}w(\bar{x})}{|x - \bar{x}|^{\alpha}} \leq C.$$

显然

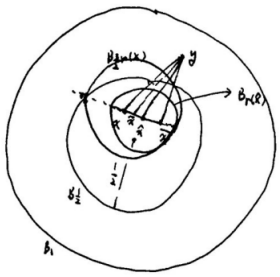
$$D_{ij}w(x) = \int_{B_1} D_{ij}\Gamma(x-y)(f(y) - f(x))dy - f(x)\int_{\partial B_1} D_i\Gamma(x-y)\cdot\nu_j(y)dS_y$$

$$D_{ij}w(\bar{x}) = \int_{B_1} D_{ij}\Gamma(\bar{x}-y)(f(y) - f(\bar{x}))dy - f(\bar{x})\int_{\partial B_1} D_i\Gamma(\bar{x}-y)\cdot\nu_j(y)dS_y$$

<u>若$r \geq \frac{1}{4}$.</u>
显然有

$$\frac{|D_{ij}w(x) - D_{ij}w(\bar{x})|}{|x - \bar{x}|^{\alpha}} \leq 4^{\alpha}(|D_{ij}w(x)| + |D_{ij}w(\bar{x})|) \leq C([f]_{\alpha} + |f(x)| + |f(\bar{x})|).$$

<u>若$r < \frac{1}{4}$.</u>
令$\hat{x} = \frac{x + \bar{x}}{2}$.将$D_{ij}w(\bar{x}) - D_{ij}w(x)$写为



$$D_{ij}w(\bar{x}) - D_{ij}w(x) = I_1 + I_2 + I_3 + I_4 + I_5 + I_6,$$

其中

$$I_1 = f(x)\int_{\partial B_1}(D_i\Gamma(x-y) - D_i\Gamma(\bar{x}-y))\nu_j(y)dS_y$$

$$I_2 = (f(x) - f(\bar{x}))\int_{\partial B_1} D_i\Gamma(\bar{x}-y)\nu_j(y)dS_y$$

$$I_3 = \int_{B_r(\hat{x})} D_{ij}\Gamma(x-y)(f(x) - f(y))dy$$

$$I_4 = \int_{B_r(\hat{x})} D_{ij}\Gamma(\bar{x}-y)(f(y) - f(\bar{x}))dy$$

$$I_5 = (f(x) - f(\bar{x}))\int_{B_1\setminus B_r(\hat{x})} D_{ij}\Gamma(x-y)dy$$

$$I_6 = \int_{B_1\setminus B_r(\hat{x})}(D_{ij}\Gamma(x-y) - D_{ij}\Gamma(\bar{x}-y))(f(\bar{x}) - f(y))dy$$

♠ 计算.
<u>I_1.</u>

$$|I_1| = \left|f(x)\int_{\partial B_1}(D_i\Gamma(x-y) - D_i\Gamma(\bar{x}-y))\nu_j(y)dS_y\right|$$

$$
\begin{aligned}
&\le |f(x)| \int_{\partial B_1} |D_i\Gamma(x-y) - D_i\Gamma(\bar{x}-y)| dS_y \\
&\le |f(x)| \int_{\partial B_1} |D(D_i\Gamma(x-y+\theta(\bar{x}-y-x+y)))||\bar{x}-y-x+y| dS_y \\
&\le |f(x)| \int_{\partial B_1} |D(D_i\Gamma(\theta\bar{x}+(1-\theta)x-y))||\bar{x}-x| dS_y \\
&\le |f(x)| \int_{\partial B_1} |D(D_i\Gamma(\tilde{x}-y))||x-\bar{x}| dS_y \\
&\le C|f(x)| \cdot |x-\bar{x}| \int_{\partial B_1} |\tilde{x}-y|^{-n} dS_y \\
&\le C|f(x)||x-\bar{x}| \le C|f(x)| \cdot |x-\bar{x}|^{\alpha} \cdot |x-\bar{x}|^{1-\alpha} \\
&\le C|f(x)| \cdot |x-\bar{x}|^{\alpha} \cdot \left(\frac{1}{4}\right)^{1-\alpha} \\
&\le C|f(x)| \cdot |x-\bar{x}|^{\alpha}.
\end{aligned}
$$

<u>I_2.</u>

$$
\begin{aligned}
|I_2| &= \left| (f(x)-f(\bar{x})) \int_{\partial B_1} D_i\Gamma(\bar{x}-y)\nu_j(y) dS_y \right| \\
&\le [f]_\alpha \cdot |x-\bar{x}|^{\alpha} \cdot \int_{\partial B_1} |D_i\Gamma(\bar{x}-y)| dS_y \\
&\le [f]_\alpha \cdot |x-\bar{x}|^{\alpha} \cdot \int_{\partial B_1} |\bar{x}-y|^{1-n} dS_y \\
&\le [f]_\alpha \cdot |x-\bar{x}|^{\alpha} \cdot |\partial B_1| \left(\frac{1}{2}\right)^{1-n} \\
&\le C[f]_\alpha \cdot |x-\bar{x}|^{\alpha}.
\end{aligned}
$$

<u>I_3.</u>

$$
\begin{aligned}
|I_3| &= \left| \int_{B_r(\hat{x})} D_{ij}\Gamma(x-y)(f(x)-f(y)) dy \right| \\
&\le \int_{B_{3/2r}(x)} |D_{ij}\Gamma(x-y)| \cdot [f]_\alpha \cdot |x-y|^{\alpha} dy \\
&\le C[f]_\alpha \cdot \int_{B_{3/2r}(x)} |x-y|^{\alpha-n} dy \\
&= C[f]_\alpha \cdot \int_0^{3/2r} t^{\alpha-n} \cdot \omega_n \cdot t^{n-1} dt \\
&= C[f]_\alpha \cdot \omega_n \cdot \frac{1}{\alpha} \cdot \left(\frac{3}{2}r\right)^{\alpha} \\
&= C[f]_\alpha \cdot |x-\bar{x}|^{\alpha}.
\end{aligned}
$$

<u>I_4.</u>
类似于I_3，有

$$
|I_4| \le C[f]_\alpha \cdot |x-\bar{x}|^{\alpha}.
$$

$\underline{I_5}$.

$$
\begin{aligned}
|I_5| &= \left|(f(x)-f(\bar{x}))\int_{B_1\setminus B_r(\hat{x})} D_{ij}\Gamma(x-y)dy\right| \\
&\le [f]_\alpha\cdot|x-\bar{x}|^\alpha\left|\int_{\partial(B_1\setminus B_r(\hat{x}))} D_i\Gamma(x-y)\cdot\nu_j(y)dS_y\right| \\
&\le [f]_\alpha\cdot|x-\bar{x}|^\alpha\left(\int_{\partial B_1}|D_i\Gamma(x-y)|dS_y+\int_{\partial B_r(\hat{x})}|D_i\Gamma(x-y)|dS_y\right) \\
&\le [f]_\alpha\cdot|x-\bar{x}|^\alpha\left(\int_{\partial B_1}c|x-y|^{1-n}dS_y+\int_{\partial B_r(\hat{x})}c|x-y|^{1-n}dS_y\right) \\
&\le [f]_\alpha\cdot|x-\bar{x}|^\alpha\left(\int_{\partial B_1}c\left(\frac{1}{2}\right)^{1-n}dS_y+\int_{\partial B_r(\hat{x})}c\left(\frac{r}{2}\right)^{1-n}dS_y\right) \\
&\le C[f]_\alpha\cdot|x-\bar{x}|^\alpha.
\end{aligned}
$$

$\underline{I_6}$.

$$
\begin{aligned}
|I_6| &= \left|\int_{B_1\setminus B_r(\hat{x})}(D_{ij}\Gamma(x-y)-D_{ij}\Gamma(\bar{x}-y))(f(\bar{x})-f(y))dy\right| \\
&\le \int_{B_1\setminus B_r(\hat{x})}|D_{ij}\Gamma(x-y)-D_{ij}\Gamma(\bar{x}-y)||f(x\bar{x})-f(y)|dy \\
&\le \int_{B_1\setminus B_r(\hat{x})}|D(D_{ij}\Gamma(\tilde{x}-y))||x-\bar{x}|\cdot[f]_\alpha\cdot|\bar{x}-y|^\alpha dy \\
&\le \int_{B_1\setminus B_r(\hat{x})}C\cdot|\tilde{x}-y|^{-n-1}\cdot|x-\bar{x}|\cdot[f]_\alpha\cdot|\bar{x}-y|^\alpha dy \\
&= C[f]_\alpha\cdot|x-\bar{x}|\cdot\int_{B_1\setminus B_r(\hat{x})}|\tilde{x}-y|^{-n-1}|\bar{x}-y|^\alpha dy \\
&\le C[f]_\alpha\cdot|x-\bar{x}|\cdot\int_{|y-\hat{x}|\ge r}\left(\frac{1}{2}|\hat{x}-y|\right)^{-n-1}(2|\hat{x}-y|)^\alpha dy \\
&= C[f]_\alpha\cdot|x-\bar{x}|\cdot 2^{n+1+\alpha}\cdot\int_{|y-\hat{x}|\ge r}|\hat{x}-y|^{\alpha-n-1}dy \\
&= C[f]_\alpha\cdot|x-\bar{x}|\int_r^{+\infty}t^{\alpha-n-1}\cdot\omega_n\cdot t^{n-1}dt \\
&\le C[f]_\alpha\cdot|x-\bar{x}|^\alpha.
\end{aligned}
$$

其中

$$
|\bar{x}-y|\le|\tilde{x}-y|+|\bar{x}-\hat{x}|\le|\hat{x}-y|+|\hat{x}-y|
$$

$$
|\hat{x}-y|\le|\tilde{x}-y|+|\hat{x}-\tilde{x}|\le|\tilde{x}-y|+|\tilde{x}-y|.
$$

综上我们得到估计：

$$
|D_{ij}w(\bar{x})-D_{ij}w(x)|\le C(|f|+[f]_\alpha)\cdot|x-\bar{x}|^\alpha.
$$

由于$D^2w\in C^\alpha$，则$w\in C^{2,\alpha}$.显然有估计

$$
|w|_{C^{2,\alpha}(B_{1/2})}\le C|f|_{C^\alpha(B_1)}.
$$

□

♣ **讨论Poisson方程的任意解的正则性.**

定理 1 设$f \in C^{\alpha}(B_1)$（$\alpha \in (0,1)$），且$|f|_{C^{\alpha}(B_1)} < \infty$.设$u \in L^{\infty}(B_1) \cap C^2(B_1)$ 为$Poisson$方程$\Delta u = f$的解.则$u \in C^{2,\alpha}$.进一步，有估计：

$$|u|_{C^{2,\alpha}(B_{1/2})} \leq C(|u|_{L^{\infty}(B_1)} + |f|_{C^{\alpha}(B_1)}).$$

证明 设w为f在B_1上的Newton位势.则有

$$|w|_{C(B_1)} \leq C|f|_{L^{\infty}(B_1)}.$$

由于$w \in C^{2,\alpha}(B_1)$为Poisson方程在B_1上的解，且有

$$|w|_{C^{2,\alpha}(B_{1/2})} \leq C|f|_{C^{\alpha}(B_1)}.$$

令$v = u - w$，则v为B_1上的调和函数，且有

$$|v|_{L^{\infty}(B_1)} \leq |u|_{L^{\infty}(B_1)} + |w|_{L^{\infty}(B_1)} \leq |u|_{L^{\infty}(B_1)} + C|f|_{L^{\infty}(B_1)}.$$

由调和函数的内部估计，有

$$|v|_{C^{2,\alpha}(B_{1/2})} \leq C|v|_{L^{\infty}(B_1)} \leq C(|u|_{L^{\infty}(B_1)} + |f|_{L^{\infty}(B_1)}).$$

从而我们有

$$\begin{aligned}
|u|_{C^{2,\alpha}(B_{1/2})} &\leq |v|_{C^{2,\alpha}(B_{1/2})} + |w|_{C^{2,\alpha}(B_{1/2})} \\
&\leq C(|u|_{L^{\infty}(B_1)} + |f|_{L^{\infty}(B_1)}) + C|f|_{C^{\alpha}(B_1)} \\
&\leq C(|u|_{L^{\infty}(B_1)} + |f|_{C^{\alpha}(B_1)}.
\end{aligned}$$

□

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